

Support Vector Machine

Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Classification

Regression

Support Vector Machine



What is support vector?

- The main goal of SVM is to maximize the margin between the two classes. The larger the margin the better the model performs on new and unseen data.
- Generally, Support Vector Machines is considered to be a classification approach, but can be employed in both types of classification and regression problems.
- It can easily handle multiple continuous and categorical variables.
- SVM constructs a hyperplane in multidimensional space to separate different classes. SVM generates optimal hyperplane in an iterative manner, which is used to minimize an error.
- The core idea of SVM is to find a maximum marginal hyperplane (MMH) that best divides the dataset into classes.
- Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It tries to find the best boundary known as hyperplane that separates different classes in the data. It is useful when you want to do binary classification like spam vs. not spam or cat vs. dog.



Past Labeled
Data



Model Training



Prediction

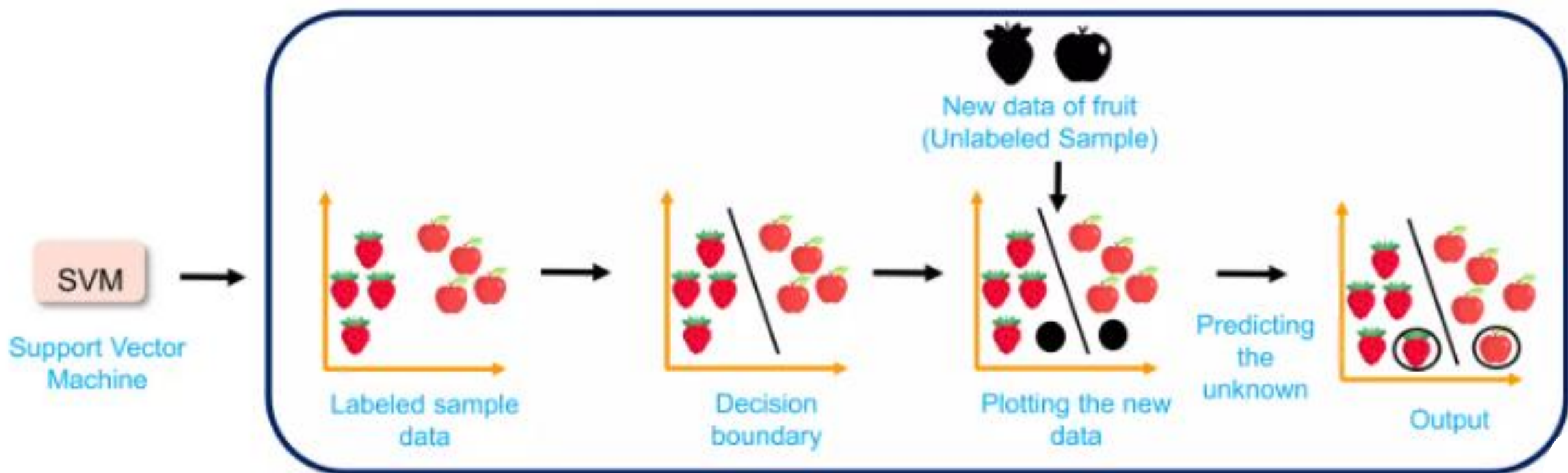


New Data

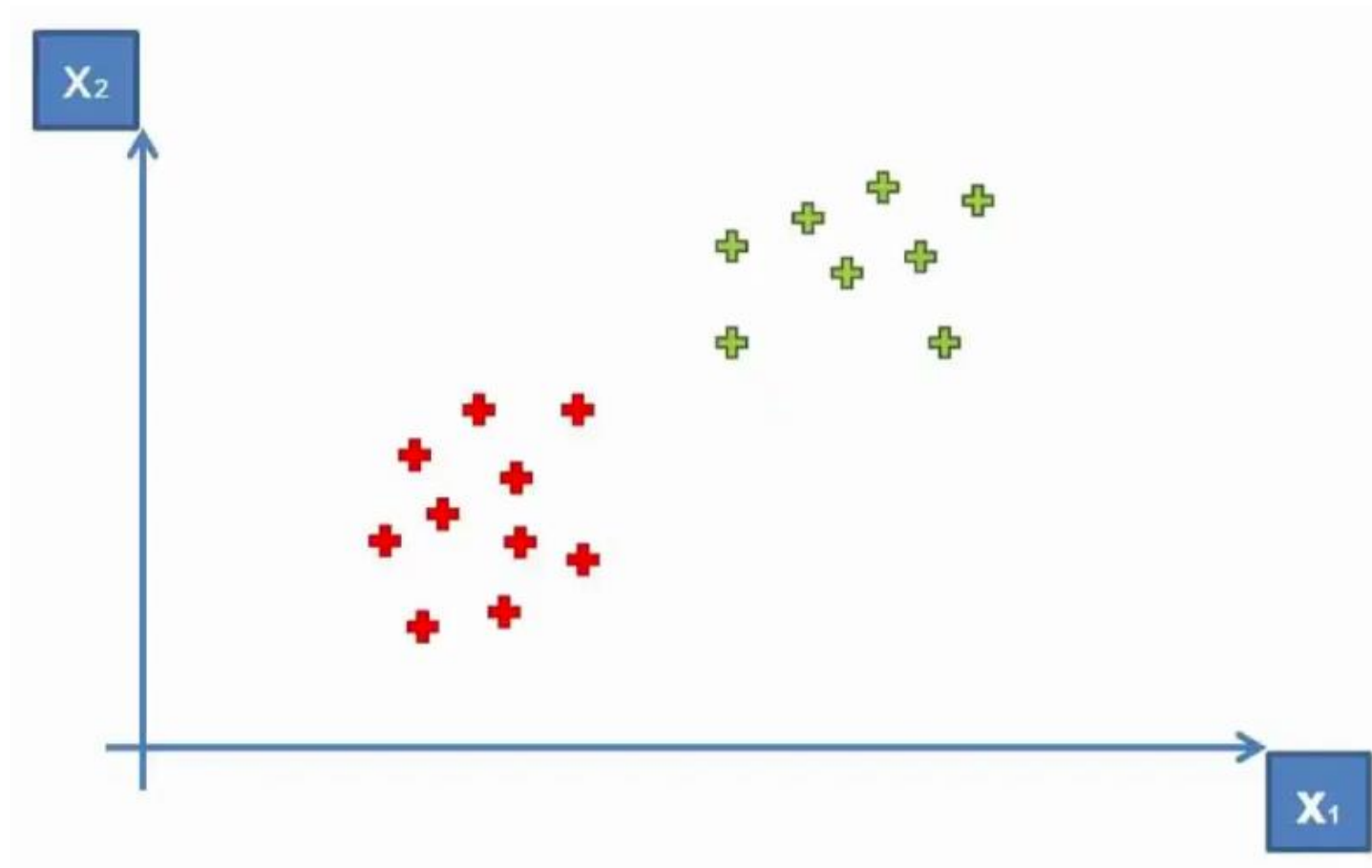


Output





Decision Vectors



Types of Support Vector Machine

Based on the nature of the decision boundary, Support Vector Machines (SVM) can be divided into two main parts:

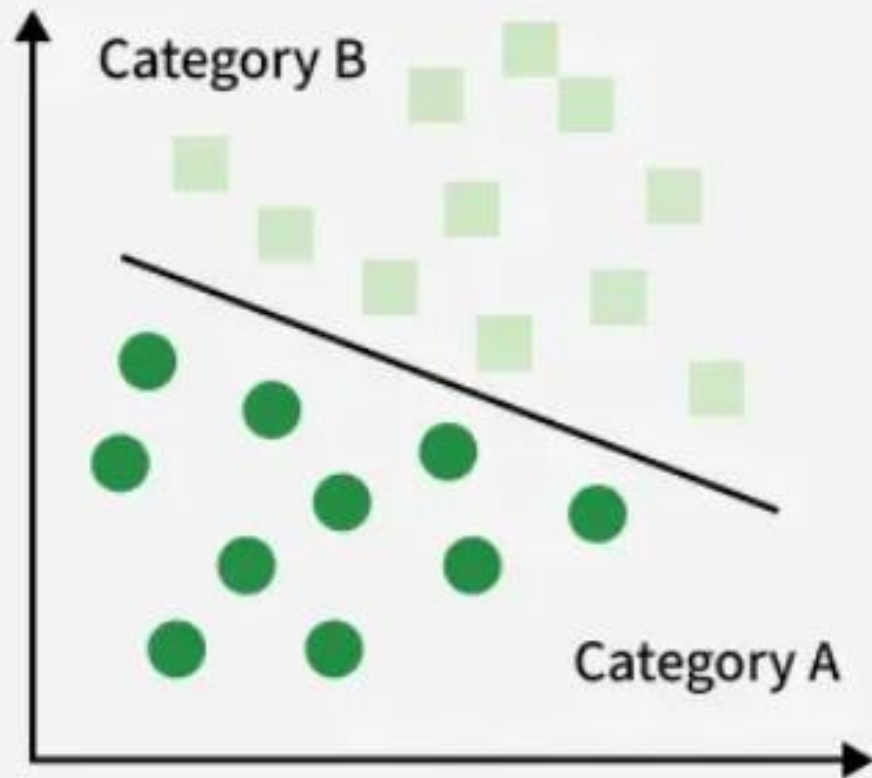
1. Linear SVM: Linear SVMs use a linear decision boundary to separate the data points of different classes.

- When the data can be precisely linearly separated, linear SVMs are very suitable.
- This means that a single straight line (in 2D) or a hyperplane (in higher dimensions) can entirely divide the data points into their respective classes.
- A hyperplane that maximizes the margin between the classes is the decision boundary.

2. Non-Linear SVM: Non-Linear SVM can be used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D).

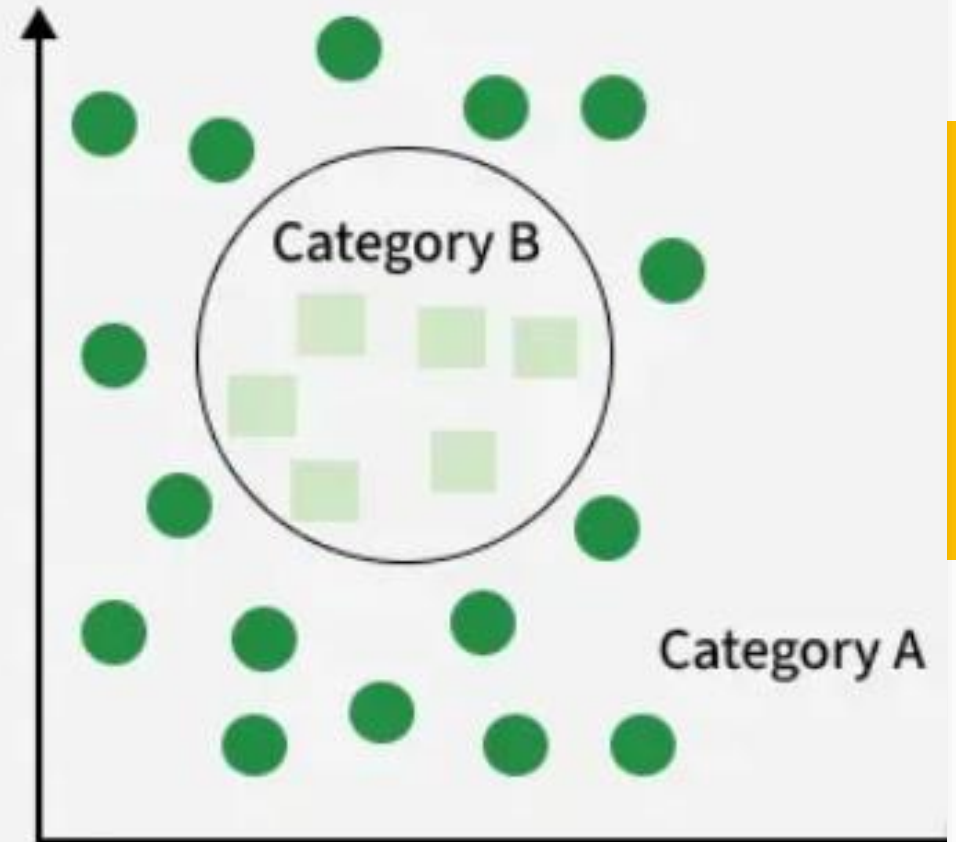
- By using kernel functions, nonlinear SVMs can handle nonlinearly separable data.
- The original input data is transformed by these kernel functions into a higher-dimensional feature space where the data points can be linearly separated.
- A linear SVM is used to locate a nonlinear decision boundary in this modified space.

Linear SVM



Vs

Non-Linear SVM

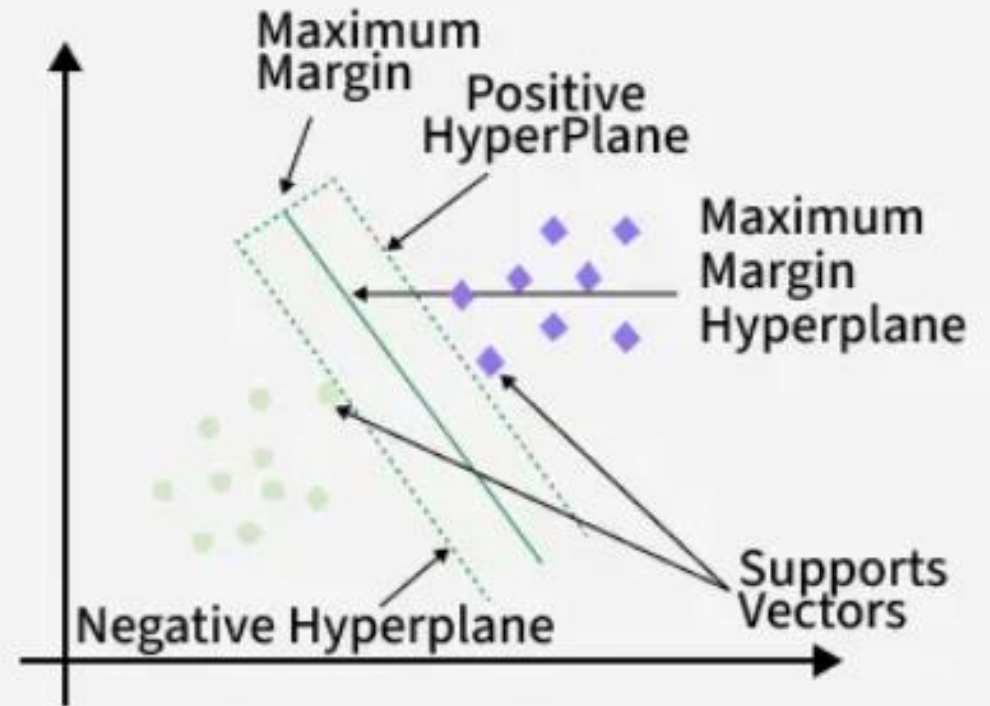


Key Concepts of Support Vector Machine

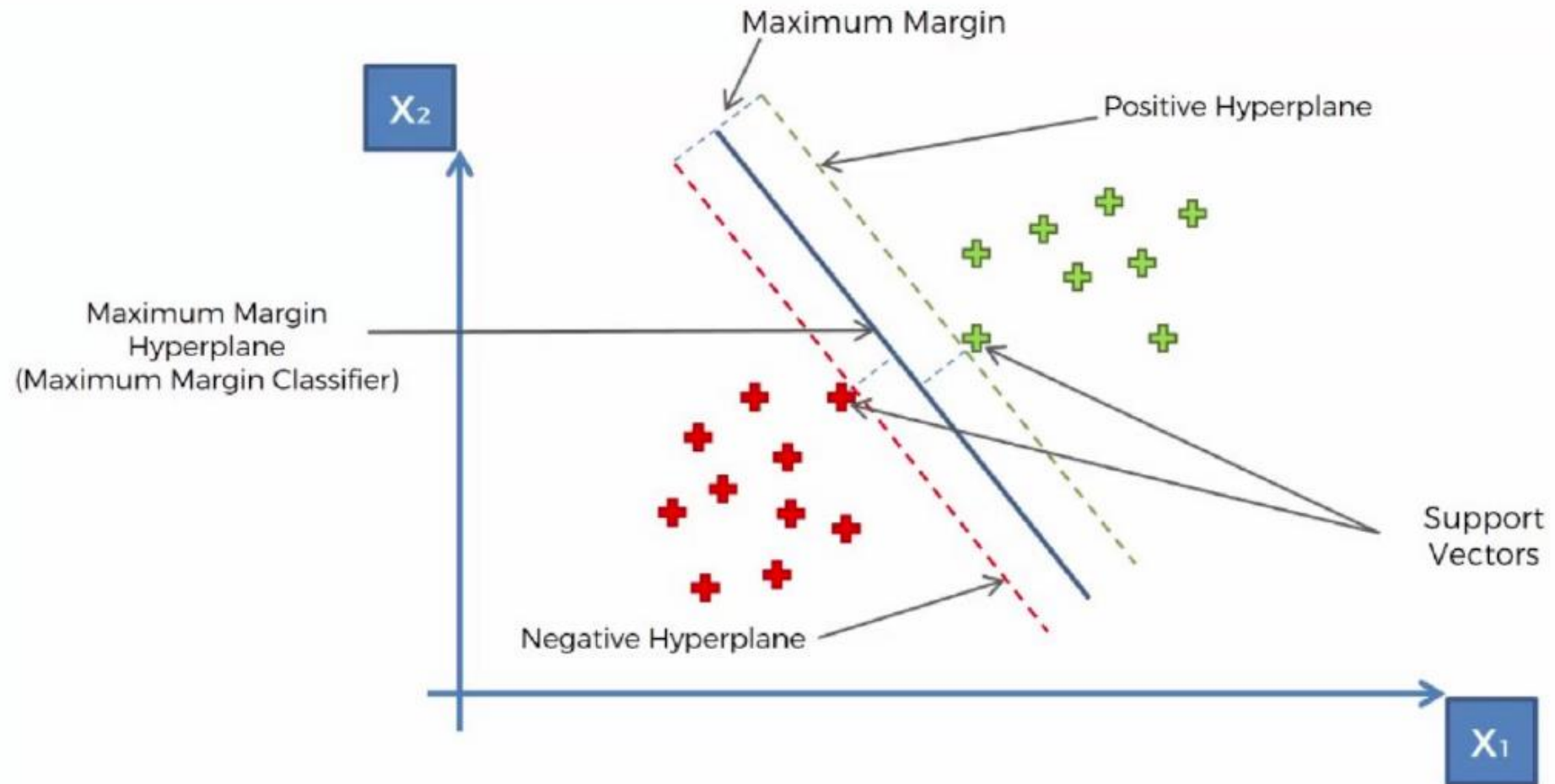
- **Hyperplane:** A decision boundary separating different classes in feature space and is represented by the equation $\mathbf{w}\mathbf{x} + \mathbf{b} = 0$ in linear classification.
- **Support Vectors:** The closest data points to the hyperplane, crucial for determining the hyperplane and margin in SVM.
- **Margin:** The distance between the hyperplane and the support vectors. SVM aims to maximize this margin for better classification performance.
- **Kernel:** A function that maps data to a higher-dimensional space enabling SVM to handle non-linearly separable data.
- **Hard Margin:** A maximum-margin hyperplane that perfectly separates the data without misclassifications.
- **Soft Margin:** Allows some misclassifications by introducing slack variables, balancing margin maximization and misclassification penalties when data is not perfectly separable.

Support Vectors & Hyperplane

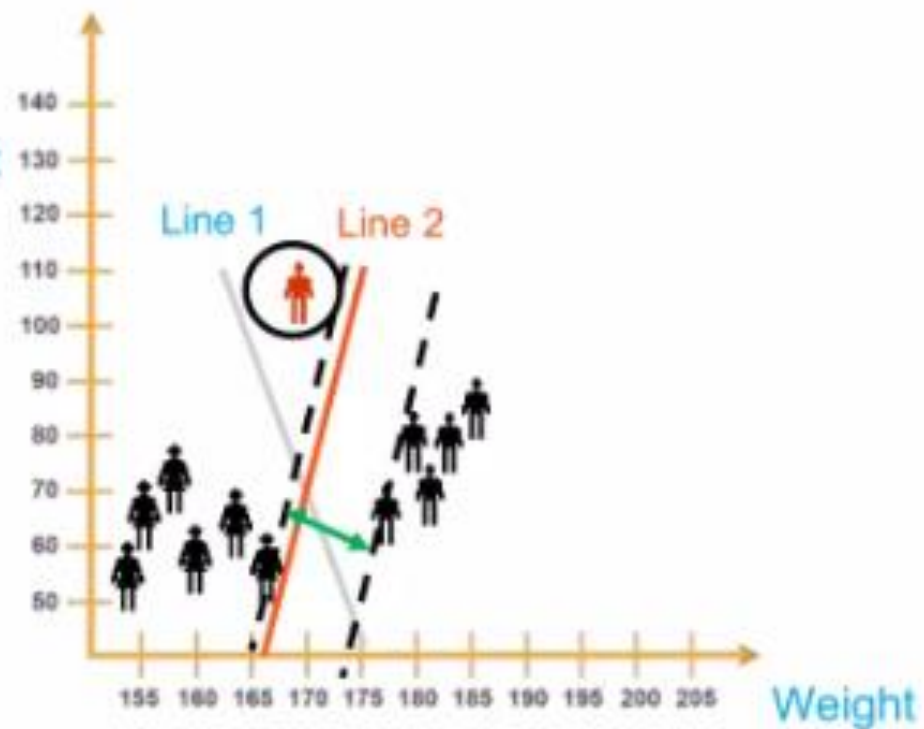
- Support Vectors are the closest data points to the hyperplane that define the class boundary.
- A hyperplane is a plane that separates different classes.



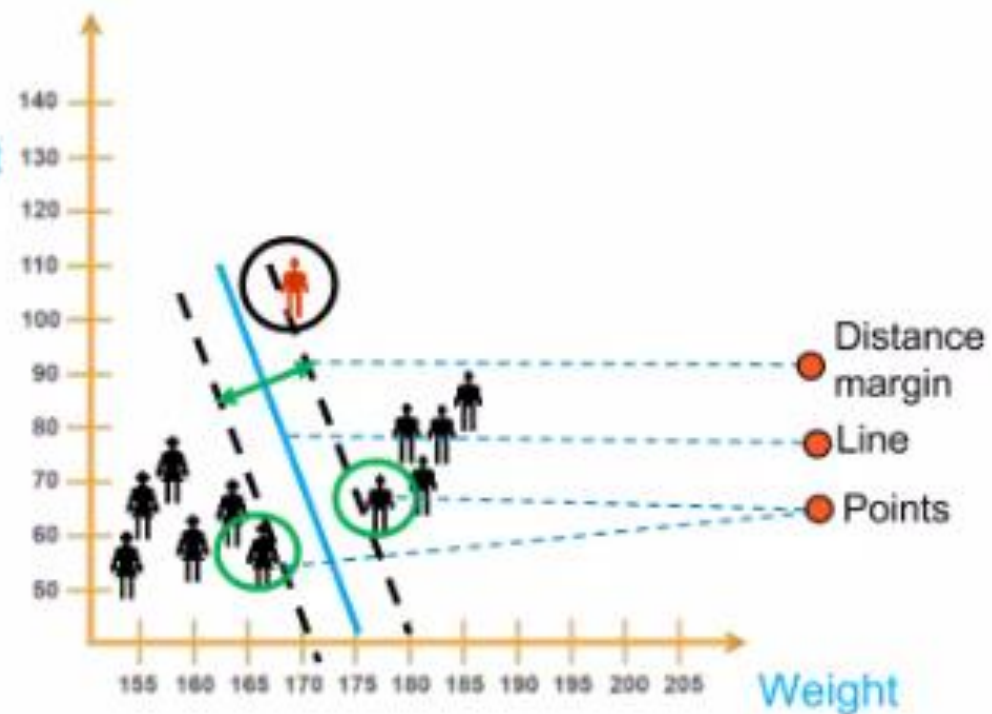
Decision Vectors



Height

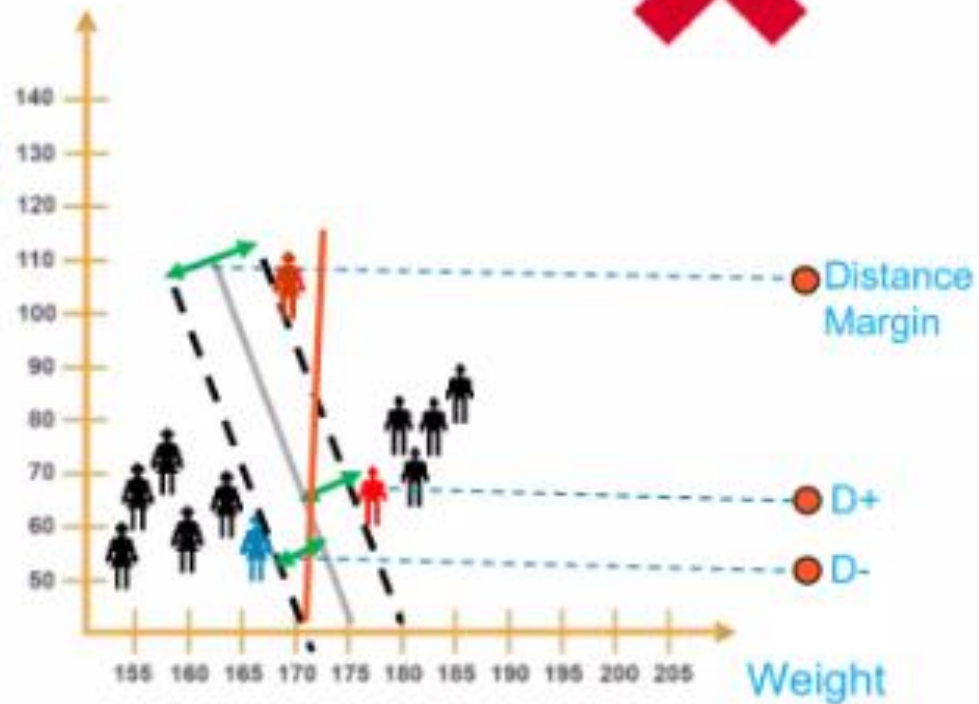


Height

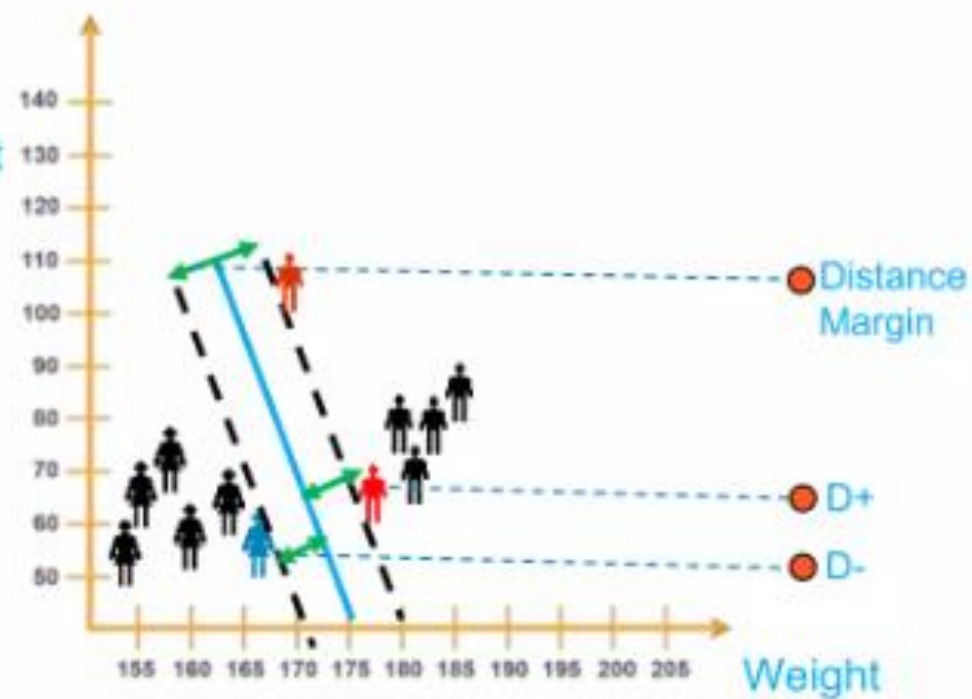




Height



Height



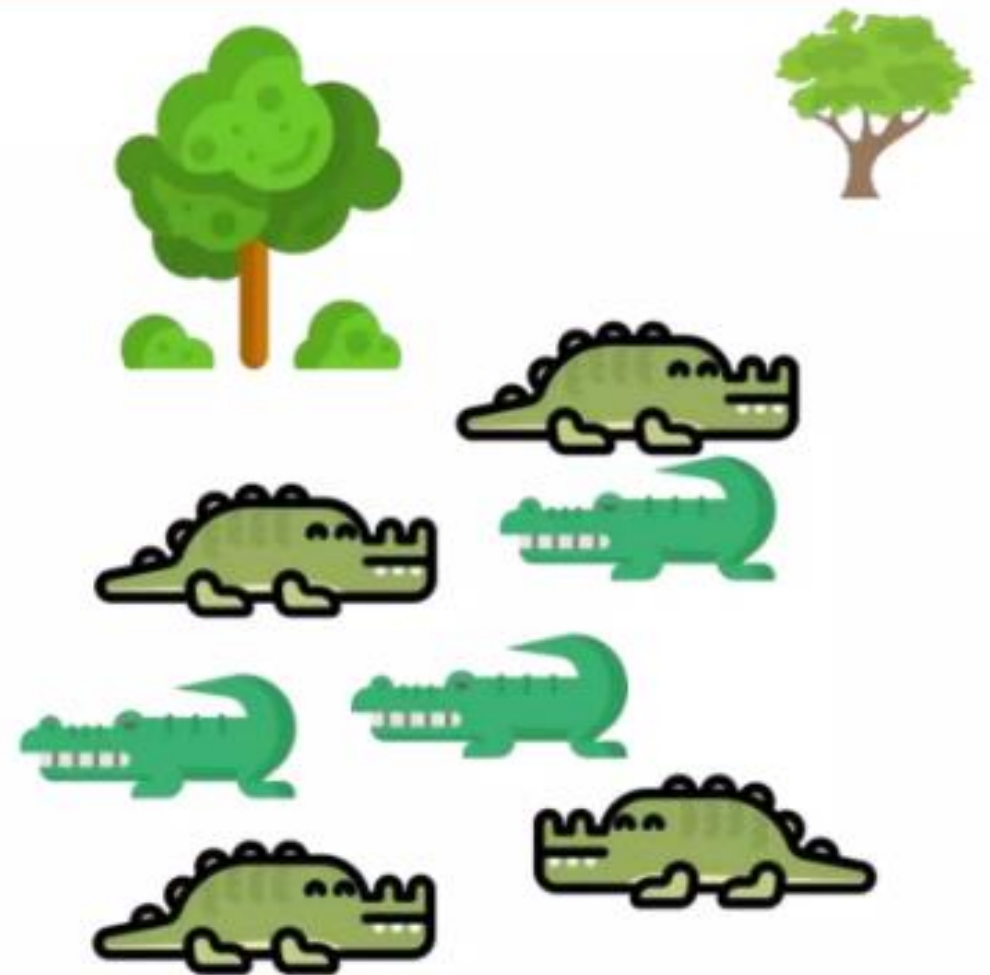
Use case – Problem Statement



Dad, is that a group of Crocodiles or Alligators?



Well that's hard to differentiate



Difference



Size

- Crocodiles are larger in size

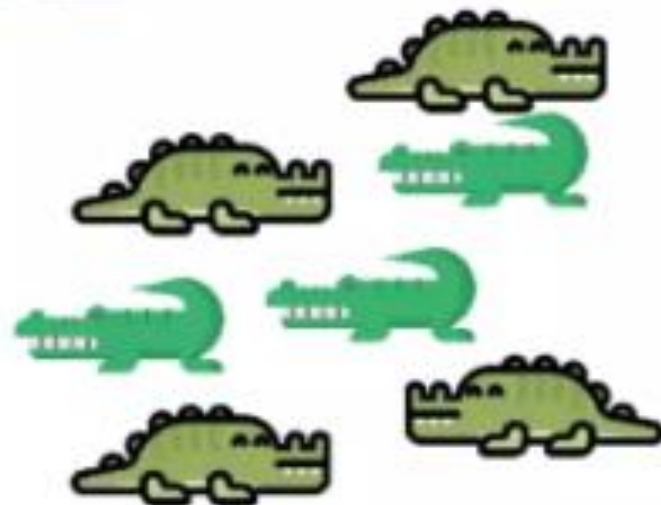


- Alligators are smaller in size

Snout Width

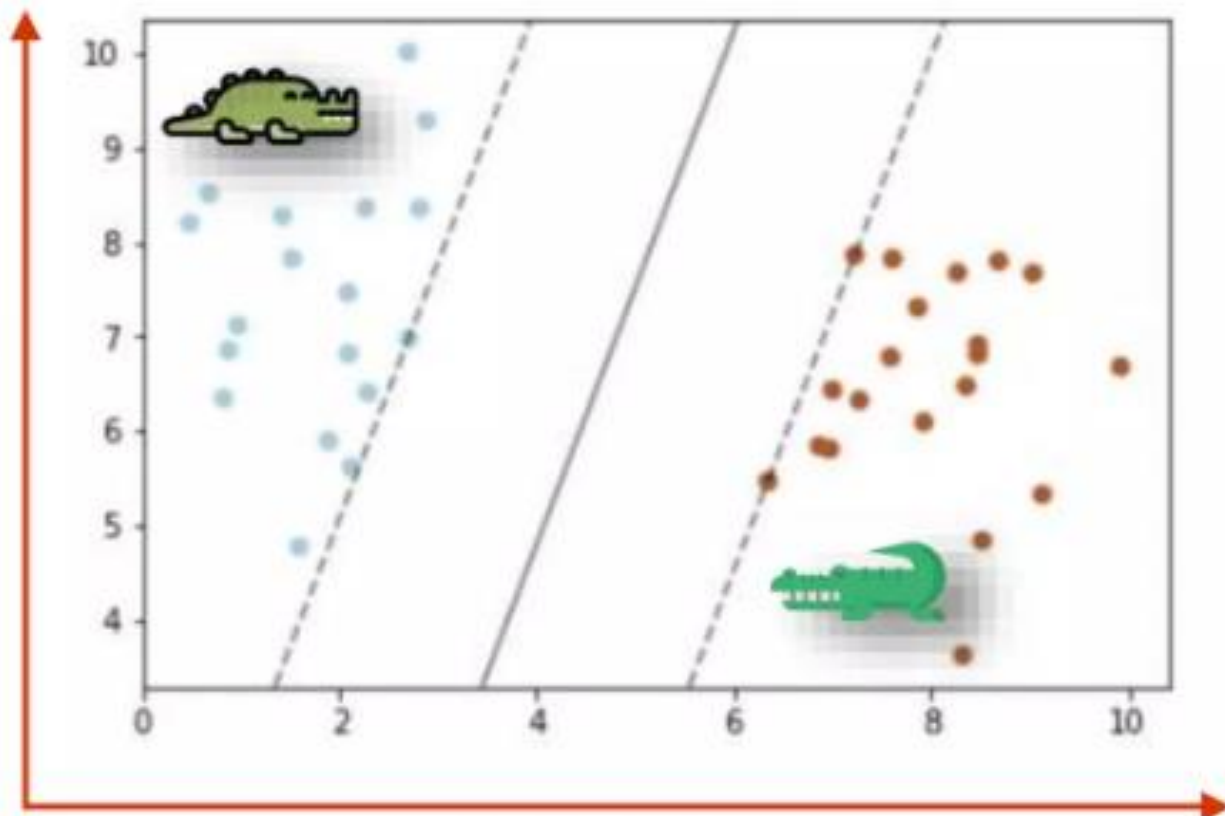
- Crocodiles have narrow snout

- Alligators have wider snout



output

Size of the
body



Width of the snout



Conclusion

Congratulations!

We have demonstrated
Support vector machine by
segregating the two classes

Where the blue data points
represents crocodiles and the
brown data points represents
alligators

The hands on example will help
you to encounter any Support
Vector Machine project in future.

What if data is not linearly separable?

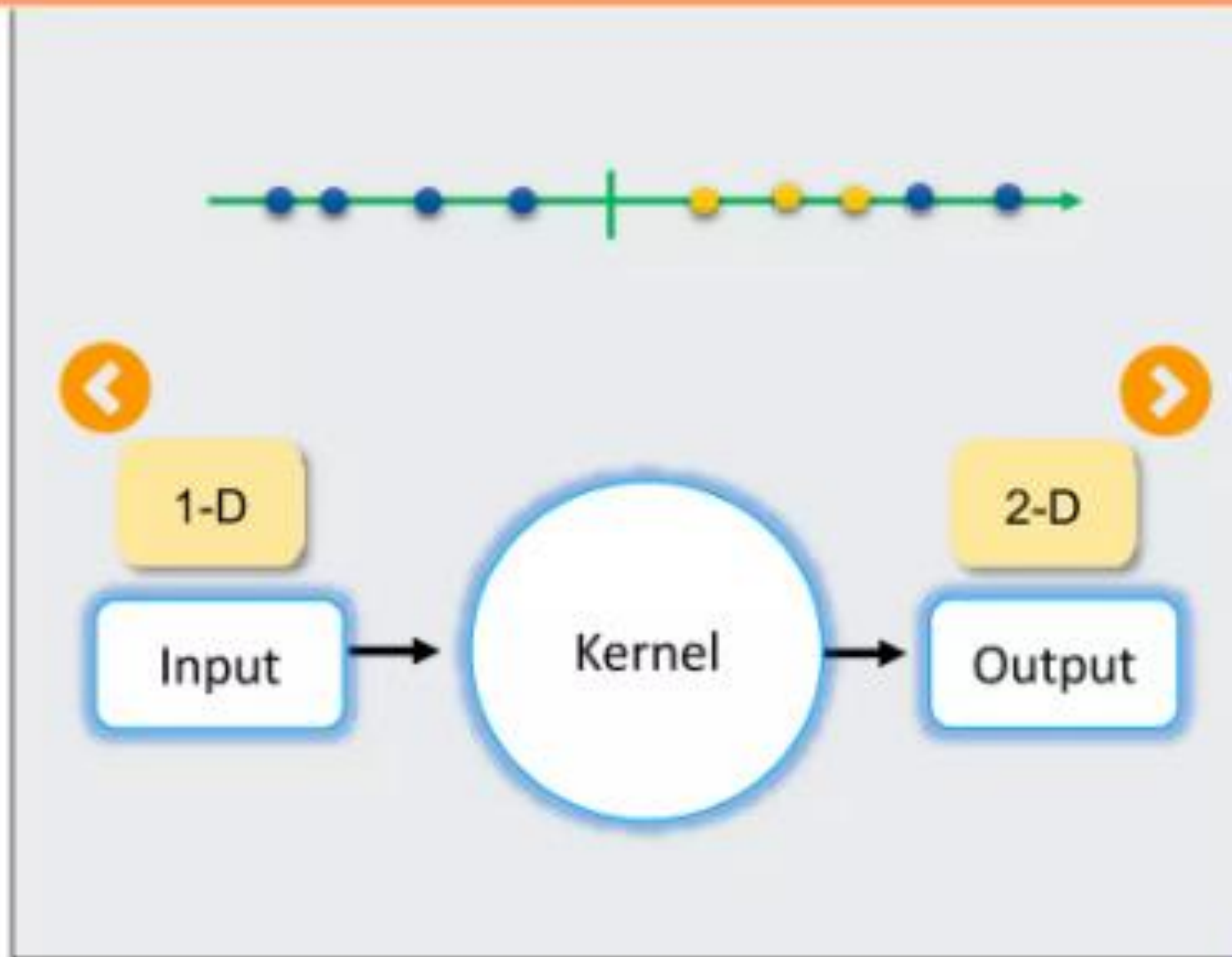
When data is not linearly separable i.e it can't be divided by a straight line, SVM uses a technique called **kernels** to map the data into a higher-dimensional space where it becomes separable. This transformation helps SVM find a decision boundary even for non-linear data.

A kernel is a function that maps data points into a higher-dimensional space without explicitly computing the coordinates in that space. This allows SVM to work efficiently with non-linear data by implicitly performing the mapping.

For example consider data points that are not linearly separable. By applying a kernel function SVM transforms the data points into a higher-dimensional space where they become linearly separable.

- **Linear Kernel:** For linear separability.
- **Polynomial Kernel:** Maps data into a polynomial space.
- **Radial Basis Function (RBF) Kernel:** Transforms data into a space based on distances between data points.

Transformation



Transformation

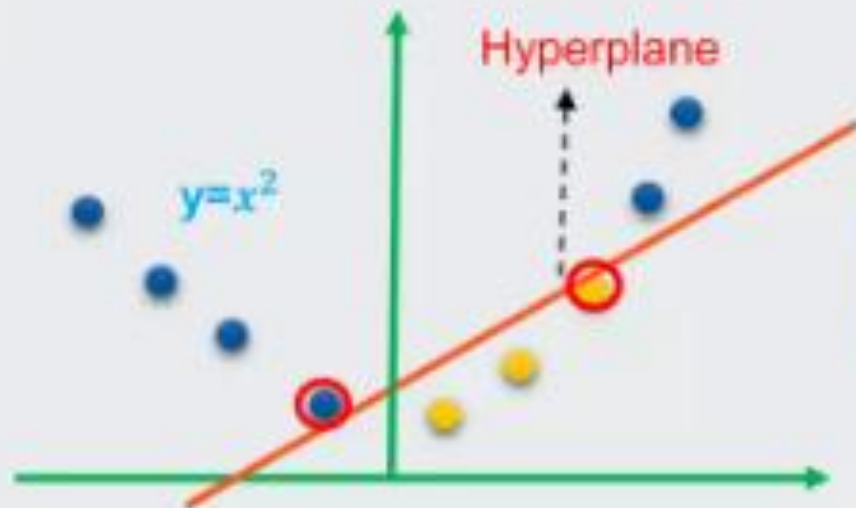
1-D

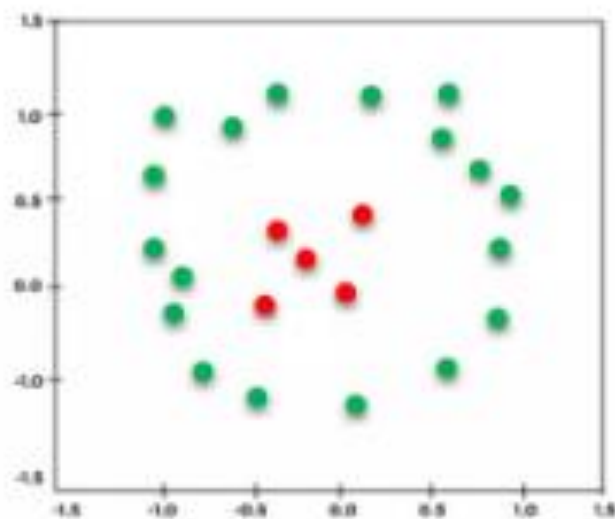


Hyperplane

$$y=x^2$$

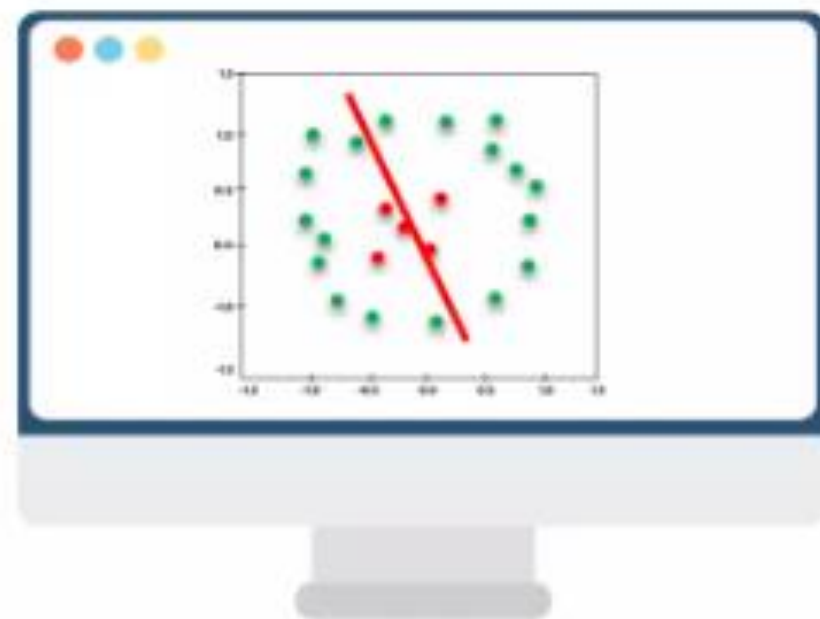
2-D



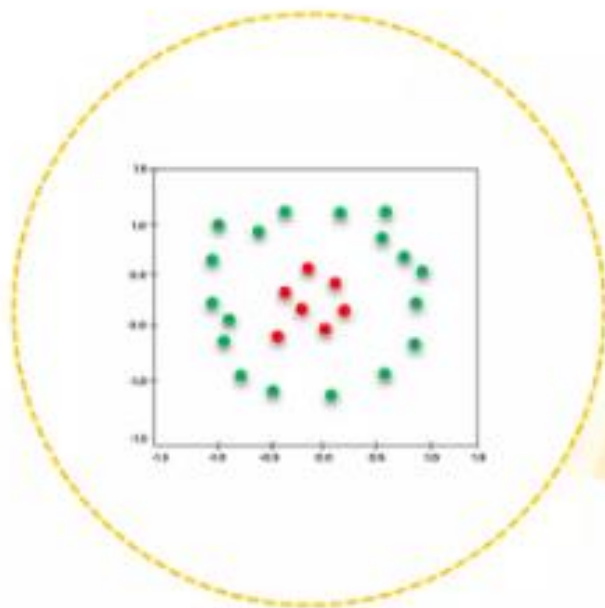


Sample Dataset

Segregate the
two classes



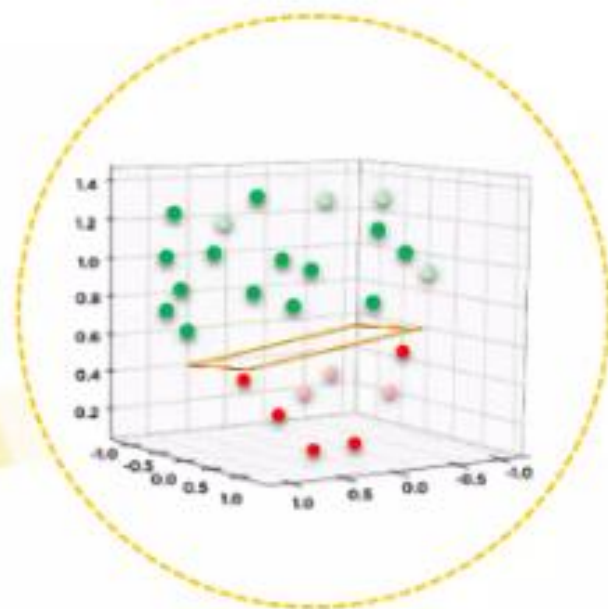
Not an optimal hyperplane



2-D



Transform



3-D

Advantages of Support Vector Machine (SVM)

- **High-Dimensional Performance:** SVM excels in high-dimensional spaces, making it suitable for image classification and gene expression analysis.
- **Nonlinear Capability:** Utilizing kernel functions like RBF and polynomial SVM effectively handles nonlinear relationships.
- **Outlier Resilience:** The soft margin feature allows SVM to ignore outliers, enhancing robustness in spam detection and anomaly detection.
- **Binary and Multiclass Support:** SVM is effective for both binary classification and multiclass classification suitable for applications in text classification.
- **Memory Efficiency:** It focuses on support vectors making it memory efficient compared to other algorithms.

Disadvantages of Support Vector Machine (SVM)

- **Slow Training:** SVM can be slow for large datasets, affecting performance in SVM in data mining tasks.
- **Parameter Tuning Difficulty:** Selecting the right kernel and adjusting parameters like C requires careful tuning, impacting SVM algorithms.
- **Noise Sensitivity:** SVM struggles with noisy datasets and overlapping classes, limiting effectiveness in real-world scenarios.
- **Limited Interpretability:** The complexity of the hyperplane in higher dimensions makes SVM less interpretable than other models.
- **Feature Scaling Sensitivity:** Proper feature scaling is essential, otherwise SVM models may perform poorly.

THANK YOU !!